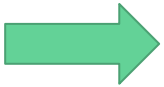
Group 2 - RNAi knockdown of *Domeless* in *Drosophila melanogaster*

Christopher Wong, Wenhao, Fuad Ahmed, Sneha Sarkar

Juvenile Hormone + Methoprene

JH is a sesquiterpenoid produced by the corpora allata, a gland in the fly's head

- Juvenile hormone (JH) controls insect development: keeps larvae in its juvenile state
  - In balance with ecdysone
- JH levels normally drop at end of 3rd instar to allow metamorphosis



Methoprene is chemically and functionally similar to JH

Juvenile hormone mimics (JHMs) like **methoprene** keep JH artificially high

- When applied at the end of the third instar, larvae exhibit a **lengthened third instar phase** (Riddiford & Ashburner, 1991)

Uses and Effects of Methoprene

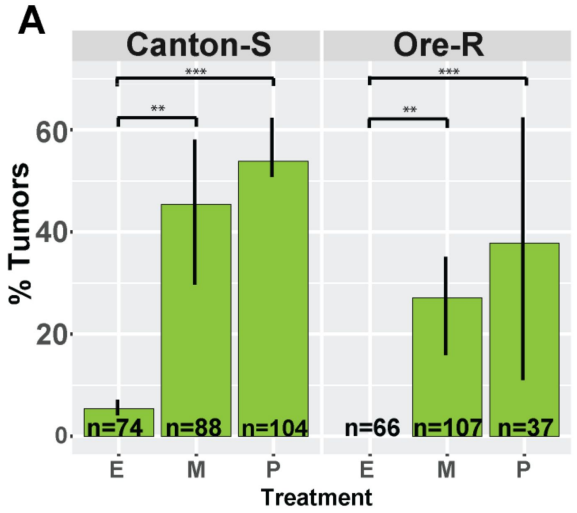
Used widely as an insecticide:

- Controls mosquito larvae (prevents spread of malaria, Zika virus)
- Protects crops from pests

Doesn’t kill; instead, impedes ability to mature

But, there’s a side effect

Methoprene and pyriproxyfen (another JHM) cause **melanotic tumor proliferation** composed of **lamellocytes** (Luu et al., 2025)



But, the specific pathways are still unclear.

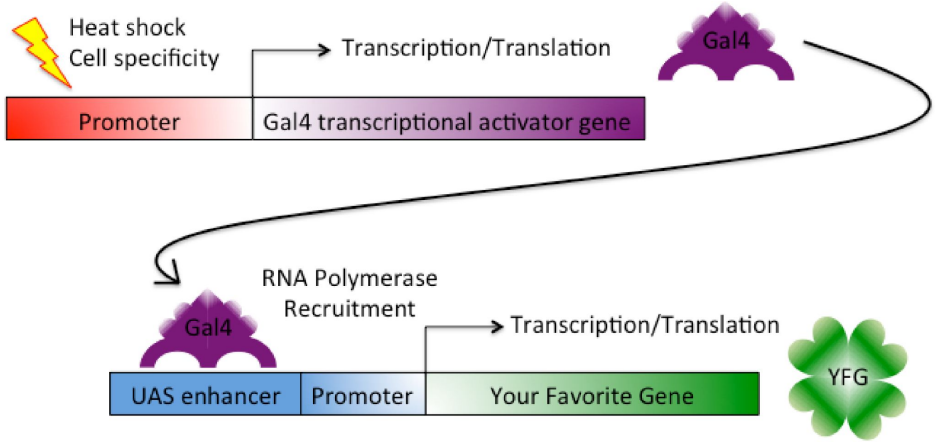
GAL4-UAS System

Two-part tool used in *Drosophila* to control gene expression in tissue-specific cells.

- GAL4 transcription factor (driver line)
  - Derived from yeast
  - Binds to specific DNA sequences ➔ increases transcription of a gene (Brand & Perrimon 1993)
  - Controlled by **tissue-specific** promoter
- UAS enhancer (responder line)
  - Placed **upstream** of the gene of interest ➔ controls expression of that gene

Transcription of UAS requires the presence of GAL4 as GAL4 binds to UAS

- Responder line flies (UAS) + driver line flies (GAL4) are crossed
- Progeny has GAL4 to **activate** UAS



Schematic of the Gal4/UAS System

Cg-GAL4

Specific driver used in *Drosophila* research used to target genes in the fly's fat body

- **Liver tissue**
- **Hemocytes**

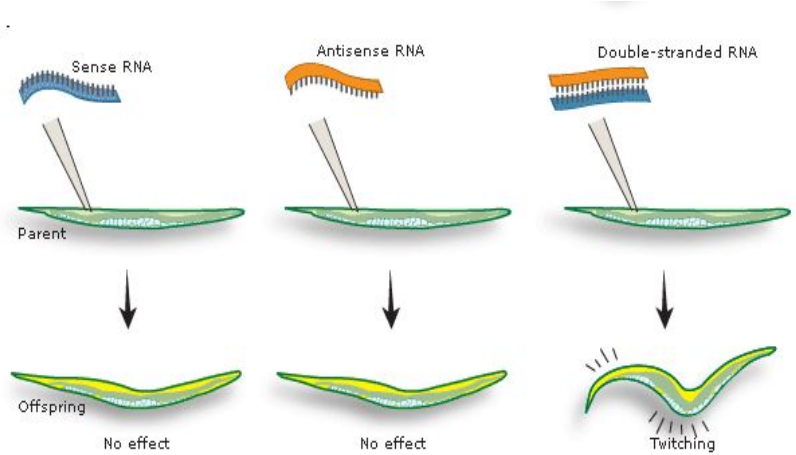


Highly active in immune responses, like JAK-STAT

Cg = collagen type IV

RNA interference

A biological phenomenon where a small double stranded RNA silences the specific corresponding gene to prevent protein production.



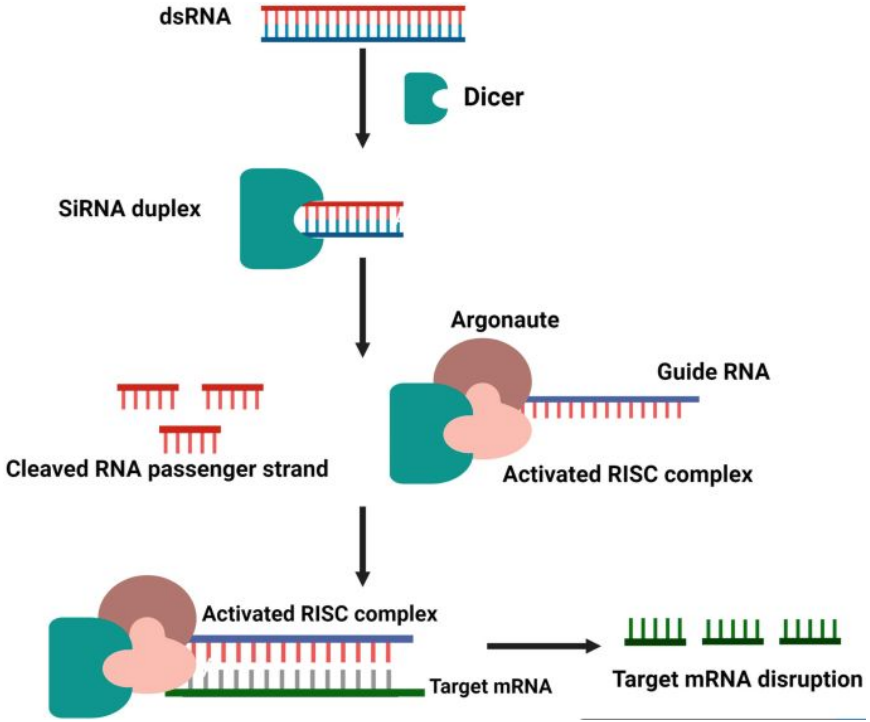
Fire and Mello experiment
Once the double-stranded RNA complementary to the unc-22 gene of the worm is injected, the result shows the silence of the target gene in worm offspring

Key finding

- The silence of the gene spread throughout the body
- This phenomenon is inherited to offsprings
- Only small dose of dsRNA can effective

RNAi mechanism

- dsRNA is injected to the cell
- Dicer “cut” dsRNA into small interfering RNA fragments (21-23 nucleotides) (Bernstein et al. 2001)
- Strand then separates to single strand and the passenger strand is degraded (Khvorova, 2003)
- Only the guide strand can enter the RNA-induced silencing complex, which silences the gene expression. (Hammond, et al 2000)

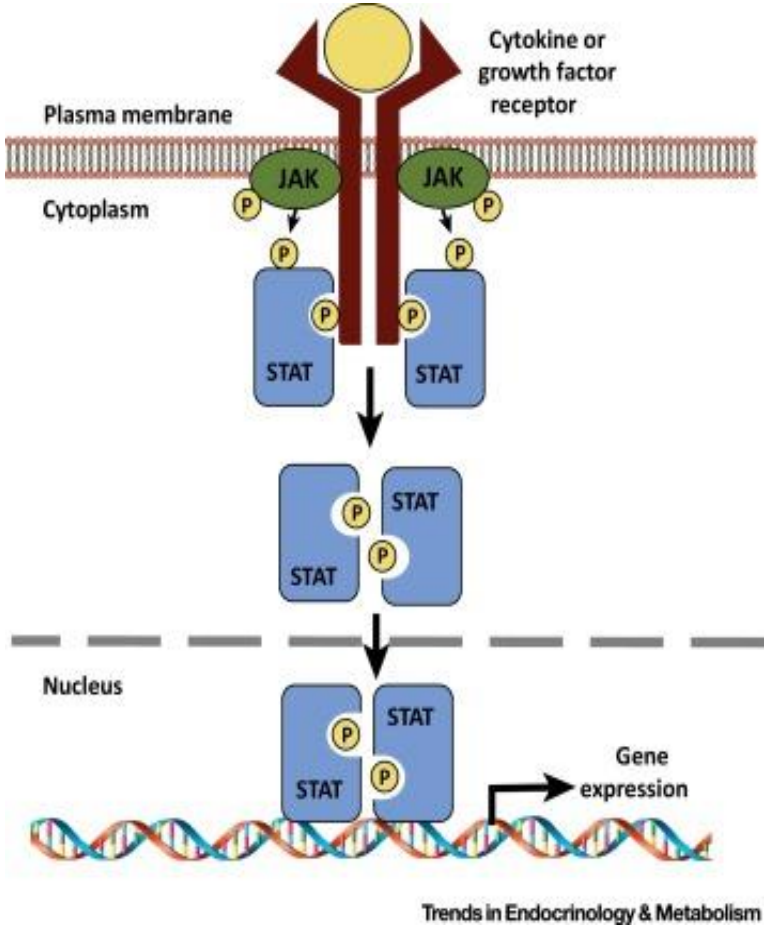


Domeless

- Belongs to cytokine receptor family
- Domeless is the only gene encoding the transmembrane receptor of the JAK-STAT pathway. (Brown, 2001)

Responsibility of Domeless

- In Lymph gland, Dome Balancing prohemocyte controlled by secretion of Upd from PSC (Posterior signaling center) (Brown, 2001)
- Dome receives the signal from Upd family to activate Hop-JAK Kinase which phosphorylate transcription factor (STAT) to control the gene expression (Flybase)



Our Hypothesis

JAK-STAT has responses ranging from proliferation and differentiation to apoptosis (Harrison, 2012)

The **JAK-STAT** pathway is also known to mediate **lamellocyte production**

- Shown by the **hop^Tum-l** mutation
- Constitutively active JAK-STAT signaling produced **melanotic tumors** (Harrison et al., 1995)

Because the JAK/STAT pathway is known to mediate lamellocyte production, supported by the hopTum-l mutation, if Domeless is knocked down by RNAi, then the amount of melanotic tumors due to JHMs such as methoprene will decrease. Without the Domeless receptor, the JAK/STAT pathway will not be able to activate and thus the formation of lamellocytes, which make up the melanotic tumors, will cease.

Experimental Design

***D. melanogaster* stocks prepared by Mr. Luu**

3 RNAi crosses with **Cg-GAL4** as the driver line:

- **Dome-RNAi**: knocks down Domeless (experimental)
- **Ebony-RNAi**: body pigmentation gene (control)
- **White-RNAi**: eye pigmentation gene (control)

Controls:

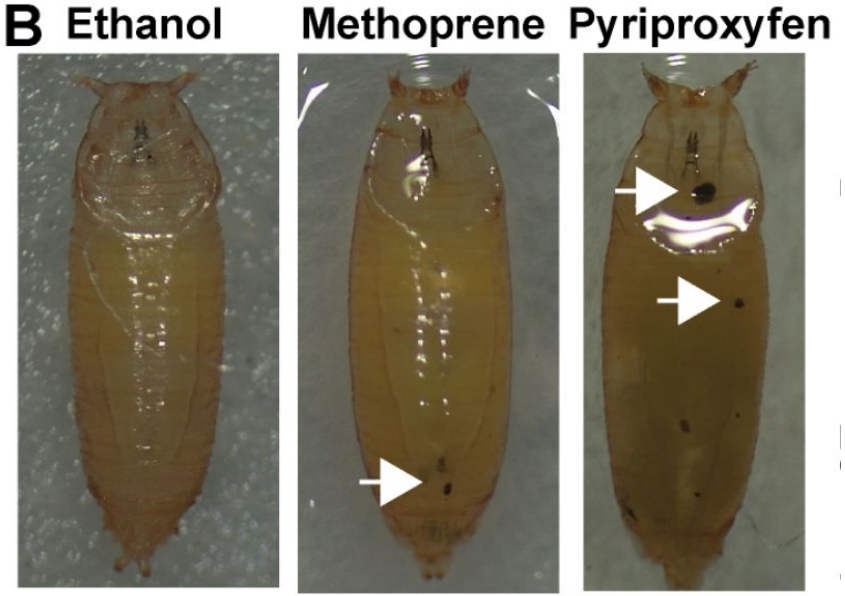
- Ethanol serves as a **vehicle control** to eliminate the possibility that ethanol (not JHM) caused tumors
- Ebony-RNAi and White-RNAi also served as **controls** because they are not involved in the JAK-STAT immune pathway

Treatments:

- 25 µL of **ethanol** (vehicle control)
- **Methoprene** at a concentration of 1 µg/µL dissolved in ethanol

Samples and Scoring

Dome RNAi + ethanol	4 vials
Dome RNAi + methoprene	10 vials
Ebony RNAi + ethanol	25 vials
Ebony RNAi + methoprene	25 vials
White RNAi + ethanol	25 vials
White RNAi + methoprene	25 vials



Scoring occurred in the prepupal stages
Tumors: Pupae was classified as either with tumors or no tumors
Viability (Done by Mr. Luu): Pupae was classified as either full or empty

Statistical Analysis

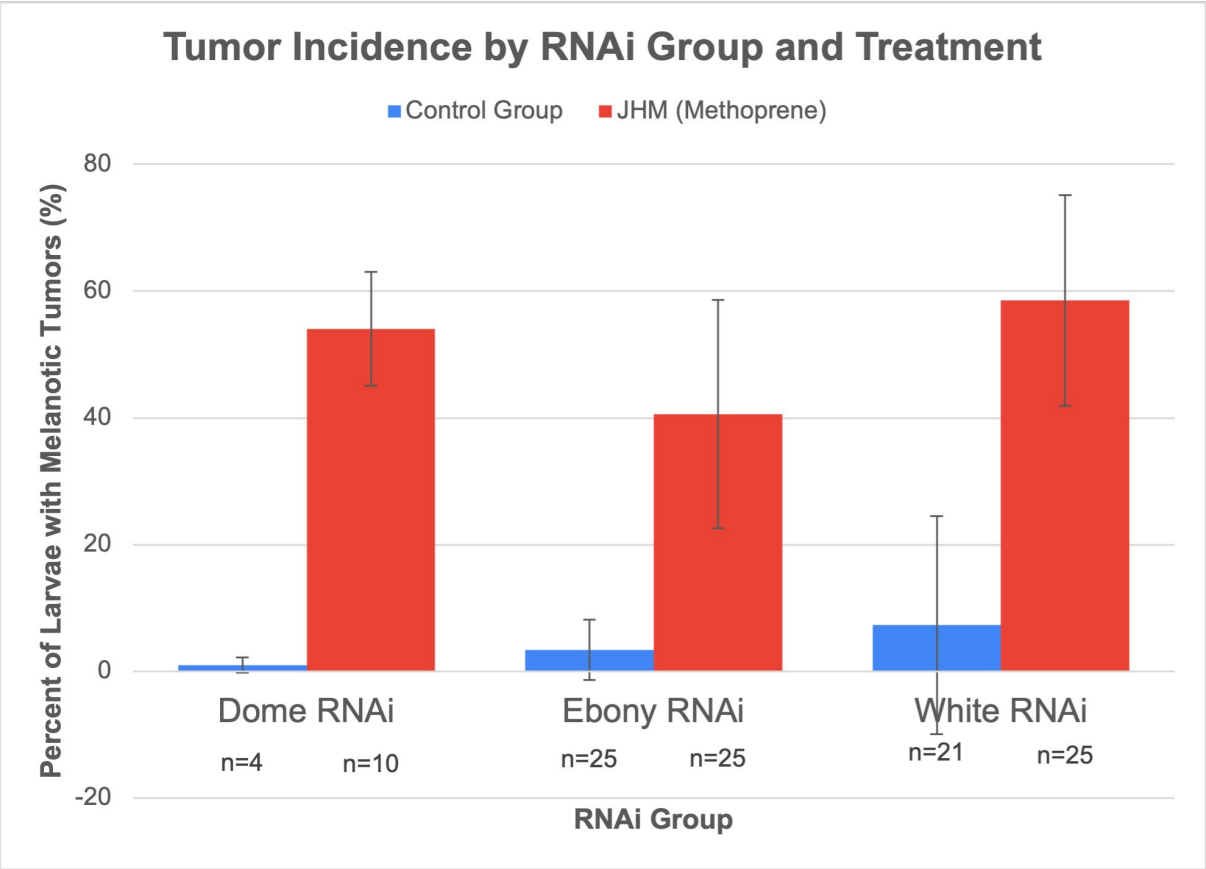
Each vial had percentages calculated:
Tumors (% of pupae with tumors) = tumors/total (tumors + no tumors)
Viability (% of pupae eclosed) = empty pupae/total (empty + full pupae)

Mean and standard deviation was calculated across all the vials for each group (using **Excel**)

Chi-square test (using Excel):

- Used on pooled pupae counts
- Compared groups in **pairs**

Results: Figure 1

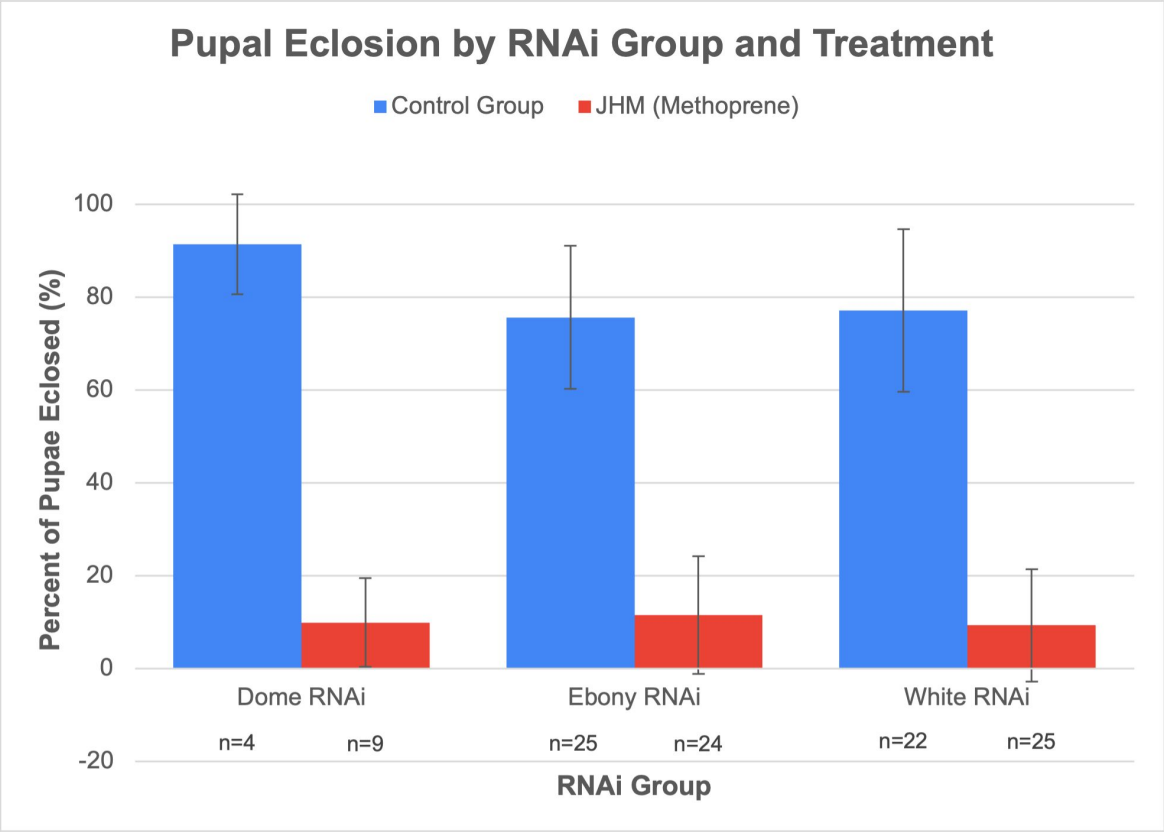


All RNAi groups formed tumors due to JHM exposure

- Dome RNAi: 54% pupae had tumors
- Ebony RNAi: 40%
- White RNAi: 58%

Control groups all had near 0%

Results: Figure 2



All RNAi groups had low viability due to JHM exposure

- Dome RNAi: 10% pupae empty
- Ebony RNAi: 11%
- White RNAi: 9%

Control groups all had near 80-90%

Chi-Square Results (Tumor) - Pairwise in JHM-treated pupae

Dome vs. Ebony

- $X^2 = 46.16$
- With a critical value of 5.99, there is a significant difference
- BUT, Ebony RNAi has lower percentage of tumors

Dome vs. White

- $X^2 = 2.72$
- With a critical value of 5.99, there is **not** a significant difference

White vs. Ebony

- $X^2 = 120.06$
- With a critical value of 5.99, there is a significant difference
- BUT, Ebony RNAi has lower percentage of tumors

Discussion

JHM induces melanotic tumor formation

- Percent of pupae with melanotic tumors of the three RNAi groups increased to 54% Dome RNAi, 40% Ebony RNAi, 58% White RNAi
- Confirms that JHM causes tumors, further supports Luu's findings

JHM decreases the viability of pupae

- Percent of pupae eclosed of the three RNAi groups decreased to 10% Dome RNAi, 11% Ebony RNAi, 9% White RNAi

Discussion

Through our chi-square test, we found that the percentage of tumors between Dome and White were not significantly different, but every comparison with Ebony was significantly different

How are our controls different from each other?

- Ebony RNAi has a much lower percentage of tumor (38%)
- White RNAi is much higher (56%)

Why?

- Ebony RNAi leads to a darker pigment
- **Melanin levels increase**

What can this cause?

- Harder to detect tumors in a darker pigment ➜ False negatives
- Directly interferes with melanin levels

Discussion

Our hypothesis was not supported.

Comparing Dome RNAi with White RNAi (better control), there is no significant difference in tumor percentage and viability

- **Suggests that Dome receptor doesn't mediate JHM-induced tumor formation**
  - **JAK-STAT is most likely not the pathway used**

Limitation

- The differences between the Chi-square test on Ebony negative control group suggest the low reliability to represent as the negative control to fully support the conclusion.
- The less amount of Dome RNAi samples also undermine the accuracy of the experiment ➔ might not fully support the conclusion.
- The single use of inducer- methoprene, could affect the generalization of the result and reduce the reliability of the experiment.

Future Direction

- Increase Dome RNAi sample size
- Studies the dose effect on methoprene
- Use different JHM to examine the accuracy
- Better control group to improve the reliability and accuracy of the experiment
- Test other pathways/other JAK-STAT components (which is what the other groups did)

Implications

- The insecticide derived from JHMs are used for mosquito and pest control
- Identifying the specific pathway ➜ identifying specific effects on mosquitoes + pest... AND other species
    - Some immune pathways are conserved
- Identifying effects on the ecosystem

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